media. However liquid media (broth) does not contain agar. Figure 5.2 shows types of media depending on physical nature.

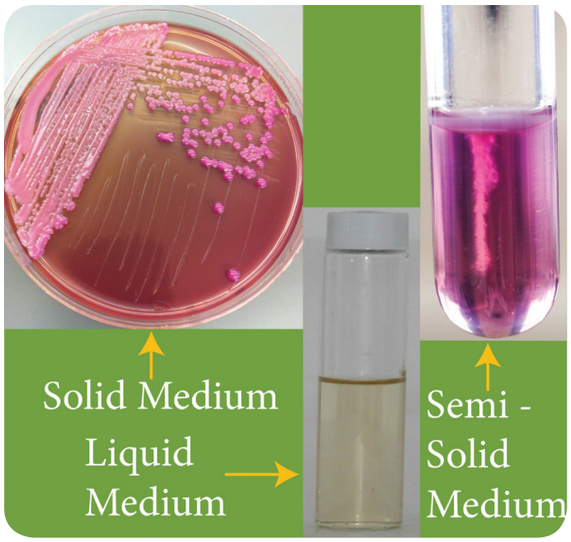


Figure 5.2: Solid, liquid and semi-solid media

5.2.2 Chemical Nature of Medium

- **Synthetic medium**

Chemically defined synthetic Medium is used for various experiments. This medium is prepared exclusively from pure substances with known chemical composition and concentrations. This is widely used in research to find the type of compound metabolized by the experimental organism.

- **Non-synthetic medium**

The medium in which the exact chemical composition and the concentration of each ingredient is not certainly known is called non-synthetic medium. In this medium, crude materials such as meat extract, yeast extract, various sugars, molasses and corn steep broth are used. This supports the growth of a variety of microorganisms. It is otherwise called as complex medium.

Infobits

Veggitone is a vegetable based product containing peptones. It is made from raw materials such as peas and fungal proteins that are digested using fungal and bacterial enzymes.

5.2.3 Special Purpose Medium

i) Basal medium

This medium promotes the growth of many types of microorganisms which do not require any special nutrient supplement. It is a routine laboratory medium with Carbon and Nitrogen sources along with some minerals. Example: Nutrient Agar or Nutrient Broth. It is also called general purpose medium. It is used for subculturing the pathogens. It is a non-selective medium, which is designed to support the growth of a wide spectrum of heterotrophic organisms. (Figure 5.3)



Figure 5.3: Growth of bacteria on Nutrient agar

ii) Enriched medium

In enriched medium, substances like blood, egg or serum are added along with the basal medium. It is used to grow fastidious organisms that are very particular in their nutritional needs. Fastidious organisms

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